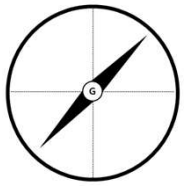
Guider: Lightweight Real-Time Performance & Fault Monitoring Framework for Embedded Linux Platforms



Peace Lee: https://www.linkedin.com/in/iipeace

Guider: https://github.com/iipeace/guider



東京 2025 LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DEC. 11-13, 2025



Contents

- Increasing complexity of Embedded Linux environments

- What is Guider?

- Real-Time Monitoring & Automation

- Remote APIs



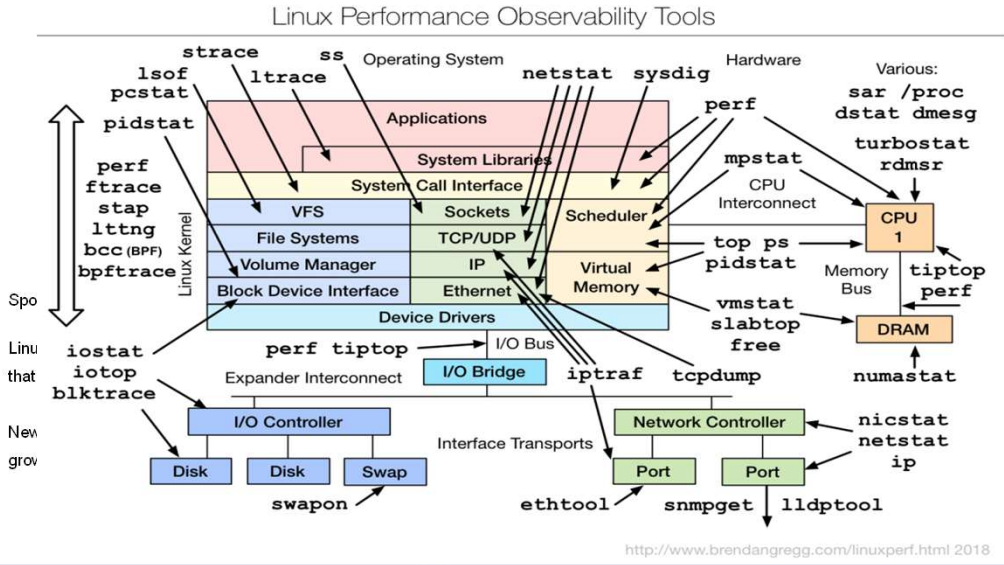
Increasing complexity of Embedded Linux environments

Challenges in:

- Capturing real-time performance issues
- Correlating logs, traces, kernel events
- Reproducing or diagnosing intermittent degradations

Limitations of existing tools:

- ftrace/perf/atrace are powerful but require manual orchestration
- Fragmented data sources
- High human overhead and lack of automation
- Perfetto (Android) is not light weight and complex





東京 2025
LINUX PLUMBERS CONFERENCE
TOKYO, JAPAN / DEC. 11-13, 2025

What is Guider?

Self-contained, open-source performance analysis tool set

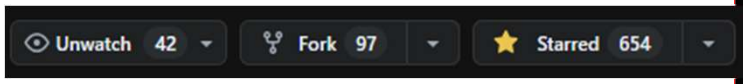
- Supports 183 commands (Monitoring, Tracing, Profiling, Utils)
- Features from System to Function, Instructions
- Leveraging ftrace, atrace, ptrace, kprobe, uprobe, DWARF, debugfs, and procfs

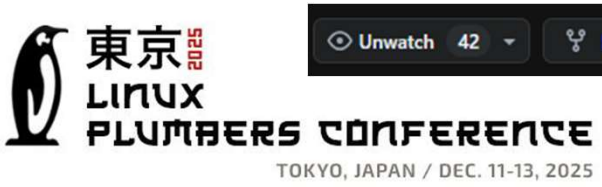
Runs on embedded Linux platforms with minimal dependencies

- Android, AGL, Tizen, webOS, custom distros
- No heavy runtime libraries or external services

Supports both command-line usage and remote API (TCP/UDS)

- Simple TCP based protocols
- Designed for multi-process and multi-device scenarios

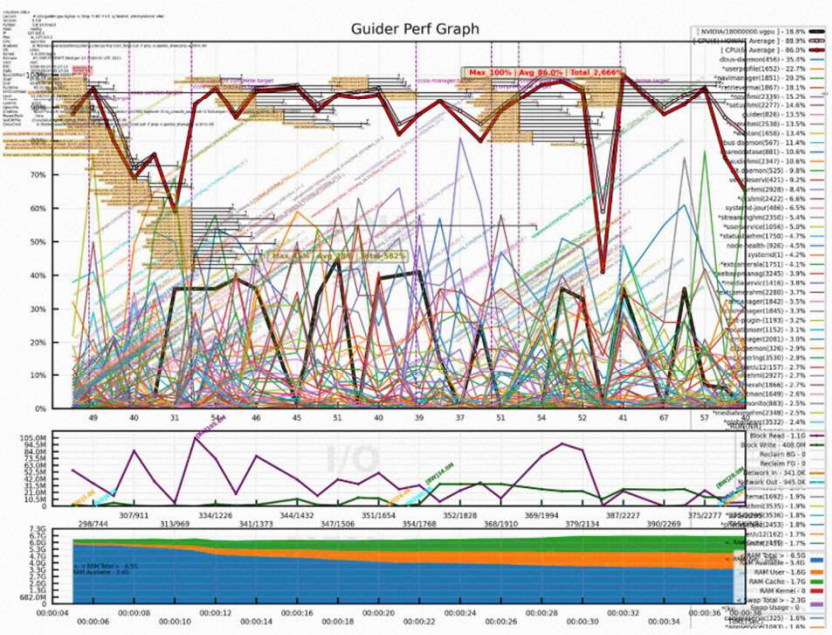






What is Guider?

Resource monitoring and visualization




```
[Top Info] [Time: 4729350.140] [Inter: 1.0] [Ctxt: 459222] [Life: +0/-0] [IRQ: 26756] [Core: 40] [Task: 553/756] [Load: 0/0/0] [RAM: 125.7G] [Swap: 8.0G]
           [N0-DMA     > diff:       0 / free:  15.5M / high:  32.0K / low:  20.0K / managed:  15.5M / min:   8.0K / present:  15.6M / spanned:  16.0M ]
           [N0-DMA32   > diff:       0 / free:   1.9G / high:   4.6M / low:   2.8M / managed:   1.9G / min: 956.0K / present:   1.9G / spanned:   4.0G ]
           [N0-Device  > diff:       0 / free:      0 / high:      0 / low:      0 / managed:      0 / min:      0 / present:      0 / spanned:      0 ]
           [N0-Movable > diff:       0 / free:      0 / high:      0 / low:      0 / managed:      0 / min:      0 / present:      0 / spanned:      0 ]
           [N0-Normal  > diff:   -7.6M / free:  99.0G / high: 318.7M / low: 191.9M / managed: 123.9G / min:  65.1M / present: 126.0G / spanned: 126.0G ]
           [IRQ > LOC: 18,929 / TIMER: 8,294 / SCHED: 7,594 / RES: 6,836 / RCU: 807 / CAL: 458 / 39: 10 / NET_RX: 10 / 16: 5 / HI: 5 / 40: 1 ]
           [Cycle: 7.0G / Inst: 3.6G / IPC: 0.51 / CacheMiss : 9.9M(30%) / BrcMiss: 14.8M(1%) / Clk: 39.0G / MinFlt: 828 / MajFlt: 0]

 ID  | CPU(Usr/Ker/Blk/IRQ)|MemAvl(Per/  User/ Cache/ Kern)| Swap( Per/ In/Out)|  PgRclm  | BlkRW | NrFlt |Blk| NrSIRQ | PgMlk | PgDirt |  Network
Total |  9 %(  1/  6/  0/  0)|126396( 98/  375/ 24856/  223)|   0(  0/  0/  0)|   0/0    |  0/0  |     0 | 0 |  17214 |  6771 |     35 |  1K/0

Core/0  | 100 %( 0 /100/ 0 / 0 )|Core/1  |   0 %( 0 / 0 / 0 / 0 )|Core/2  |   0 %( 0 / 0 / 0 / 0 )|Core/3  |   0 %( 0 / 0 / 0 / 0 )|Core/4  |   0 %( 0 / 0 / 0 / 0 )
Core/5  |   0 %( 0 / 0 / 0 / 0 )|Core/6  |   0 %( 0 / 0 / 0 / 0 )|Core/7  |   0 %( 0 / 0 / 0 / 0 )|Core/8  |  72 %( 0 /71 / 0 / 0 )|Core/9  |   0 %( 0 / 0 / 0 / 0 )
Core/10 |  15 %( 0 /14 / 0 / 0 )|Core/11 |   0 %( 0 / 0 / 0 / 0 )|Core/12 | 100 %(57 /43 / 0 / 0 )|Core/13 |   0 %( 0 / 0 / 0 / 0 )|Core/14 |   0 %( 0 / 0 / 0 / 0 )
Core/15 |   0 %( 0 / 0 / 0 / 0 )|Core/16 |  32 %( 0 /31 / 0 / 0 )|Core/17 |   3 %( 1 / 1 / 0 / 0 )|Core/18 |   0 %( 0 / 0 / 0 / 0 )|Core/19 |   0 %( 0 / 0 / 0 / 0 )
Core/20 |   0 %( 0 / 0 / 0 / 0 )|Core/21 |   0 %( 0 / 0 / 0 / 0 )|Core/22 |   0 %( 0 / 0 / 0 / 0 )|Core/23 |   0 %( 0 / 0 / 0 / 0 )|Core/24 |   0 %( 0 / 0 / 0 / 0 )
Core/25 |   0 %( 0 / 0 / 0 / 0 )|Core/26 |   0 %( 0 / 0 / 0 / 0 )|Core/27 |   0 %( 0 / 0 / 0 / 0 )|Core/28 |   0 %( 0 / 0 / 0 / 0 )|Core/29 |   0 %( 0 / 0 / 0 / 0 )
Core/30 |   0 %( 0 / 0 / 0 / 0 )|Core/31 |   0 %( 0 / 0 / 0 / 0 )|Core/32 |   0 %( 0 / 0 / 0 / 0 )|Core/33 |   0 %( 0 / 0 / 0 / 0 )|Core/34 |   0 %( 0 / 0 / 0 / 0 )
Core/35 |   0 %( 0 / 0 / 0 / 0 )|Core/36 |   0 %( 0 / 0 / 0 / 0 )|Core/37 |   0 %( 0 / 0 / 0 / 0 )|Core/38 |   0 %( 0 / 0 / 0 / 0 )|Core/39 |   0 %( 0 / 0 / 0 / 0 )

     DEV      |BUSY| AVQ |  READ  |  WRITE |   FREE(   DIFF)|USAGE|  TOTAL | NrAVF |  FS  |        MountPoint <Option>
/dev/sda2     |  0%|    0|       0|       0| 608.0M(      0)|  37%| 975.0M |  63.7K|  ext4| /boot <ro,relatime>
/dev/sda1     |  0%|    0|       0|       0| 503.0M(      0)|   1%| 510.0M |      0|  vfat| /boot/efi <rw,relatime>

          Network          |                Receive                |               Transfer
    Dev    |       IP       |  Size  | Packet | Error | Drop | Multicast |  Size  | Packet | Error | Drop | Multicast
  docker0  |   172.17.0.1   |      0 |      0 |     0 |    0 |         0 |      0 |      0 |     0 |    0 |         0
  enp7s0   | 166.104.144.86 |   4.0K |     19 |     0 |    0 |         1 |    652 |      6 |     0 |    0 |         0
     lo    |   127.0.0.1    |      0 |      0 |     0 |    0 |         0 |      0 |      0 |     0 |    0 |         0
[ TOTAL ]  |                |   4.0K |     19 |     0 |    0 |         1 |    652 |      6 |     0 |    0 |         0

  Process (   PID/  PPID/  Nr/ Pri)| CPU(Usr/Ker/Dly)| VSS( RSS/Txt/Shr/Swp)| Blk(  RD/  WR/NrFlt)| SID | USER | FD | LifeTime|      Parent
  screen (  5453/     1/   1/C  0)|  99( 57/ 42/  -)|  11(   7/  0/  2/  0)|   0(   0/   0/   0)| 5453|    -| 64|54d:05:34|       systemd(1)
                MEM(STACK/1)      | VSS: 132.0K / RSS: 116.0K / PSS: 116.0K / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT: 116.0K
                MEM(FILE/12)      | VSS:   6.2M / RSS:   2.5M / PSS: 405.0K / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT: 132.0K
                MEM(ETC/3)        | VSS:  20.0K / RSS:   4.0K / PSS:     0 / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT:     0
                MEM(ANON/11)      | VSS:   5.3M / RSS:   5.1M / PSS:   5.1M / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT:   5.1M
                MEM(SUM)          | VmPeak: 11.6M, VmHWM: 7.8M, VmData: 5.4M, HugetlbPages: 0, RssAnon: 5.4M, RssFile: 2.4M, RssShmem: 0

  yes (848357/806199/   1/C  0)|  99(  0/ 99/  -)|   5(   0/  0/  0/  0)|   0(   0/   0/   0)| 5454|    -| 64| 00:00:03|       vi(806199)
                MEM(STACK/1)      | VSS: 132.0K / RSS:  12.0K / PSS:  12.0K / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT:  12.0K
                MEM(FILE/4)       | VSS:   5.0M / RSS:   1.7M / PSS: 100.0K / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT:  40.0K
                MEM(ETC/3)        | VSS:  20.0K / RSS:   4.0K / PSS:     0 / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT:     0
                MEM(ANON/3)       | VSS: 160.0K / RSS:  44.0K / PSS:  44.0K / SWAP:     0 / HUGE:     0 / LOCK:     0 / SDRT:     0 / PDRT:  44.0K
                MEM(SUM)          | VmPeak: 5.3M, VmHWM: 520.0K, VmData: 180.0K, HugetlbPages: 0, RssAnon: 64.0K, RssFile: 456.0K, RssShmem: 0

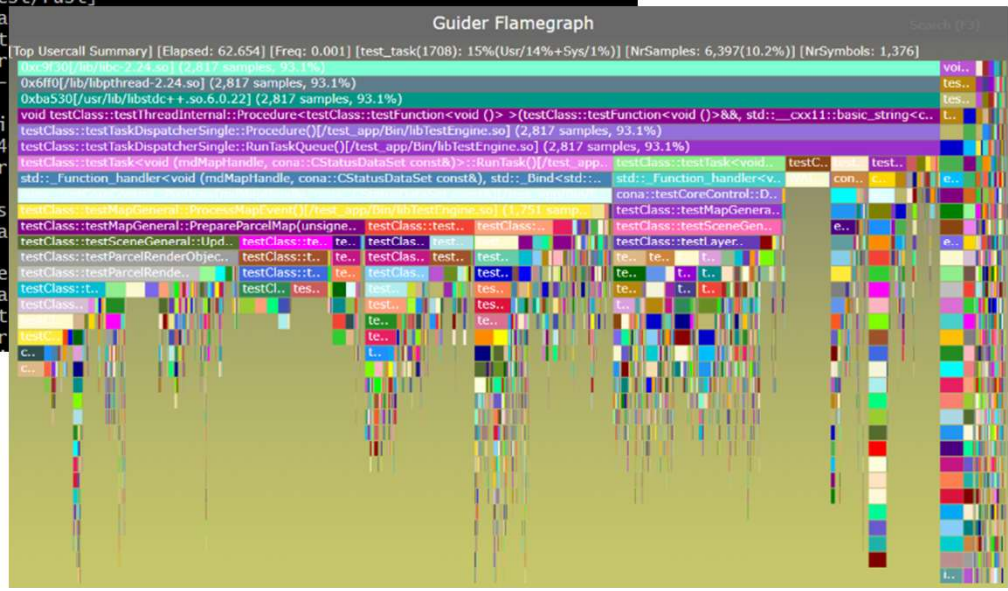
kworker/u80:3+ (848358/     2/   1/C  0)|  81(  0/ 81/  -)|   0(   0/  0/  -/  -)|   0(   0/   0/   0)|    -|    -|  -| 00:00:01|       kthreadd(2)
```



What is Guider?

Function monitoring and visualization

```
[Top Usercall Info] [Time: 281819.230] [Inter: 0.999] [Sample: 637] [SYS: 35%/54.7G] [rust(225737): 70%(U41%+S29%)/676K] [guider(225683): 31%/37M] [Freq: 1ms]
Usage  | Function [PATH] <Sample>
 87.4% | write@GLIBC_2.28 [/usr/lib/x86_64-linux-gnu/libpthread-2.31.so] <Cnt: 557>
         100.0% |  <- std::io::buffered::bufwriter::BufWriter<W>::flush_buf[/home/iipeace/work/test/rust]
                   <- _<std::io::buffered::linewritershim::LineWriterShim<W> as std::io::Write>::write_all[/home/iipeace/work/test/rust]
                   <- _<std::io::Write::write_fmt::Adapter<T> as core::fmt::Write>::write_str#0x24d40[/home/iipeace/work/test/rust]
                   <- core::fmt::write[/home/iipeace/work/test/rust]
                   <- _<&std::io::stdio::Stdout as std::io::Write>::write_fmt[/home/iipeace/work/test/rust]
                   <- std::io::stdio::_print[/home/iipeace/work/test/rust] <- rust::main[/home/iipeace/work/test/rust]
                   <- core::ops::function::FnOnce::call_once#0x81d0[/home/iipeace/work/test/rust]
                   <- std::sys_common::backtrace::__rust_begin_short_backtrace[/home/iipeace/work/test/rust]
                   <- std::rt::lang_start::_{{closure}}[/home/iipeace/work/test/rust] <- std::rt::lang_start_internal[/home/iipeace/work/test/rust]
                   <- std::rt::lang_start[/home/iipeace/work/test/rust] <- main[/home/iipeace/work/test/rust]
                   <- __libc_start_main@GLIBC_2.2.5[/usr/lib/x86_64-linux-gnu/libc-2.31.so] <- _start[/home/iipeace/work/test/rust] <Cnt: 557>
---------------------------------------------------------------------------------------------------------------------------------------------------
  2.2% | _<&std::io::stdio::Stdout as std::io::Write>::write_fmt [/home/iipeace/work/test/rust] <Cnt: 14>
         100.0% |  <- std::io::stdio::_print[/home/iipeace/work/test/rust] <- rust::main[/home/iipeace/work/test/rust]
                   <- core::ops::function::FnOnce::call_once#0x81d0[/home/iipeace/work/test/rust]
                   <- std::sys_common::backtrace::__rust_begin_short_backtrace[/home/iipeace/work/test/rust]
                   <- std::rt::lang_start::_{{closure}}[/home/iipeace/work/test/rust] <- std::rt::la
                   <- std::rt::lang_start[/home/iipeace/work/test/rust] <- main[/home/iipeace/work/t
                   <- __libc_start_main@GLIBC_2.2.5[/usr/lib/x86_64-linux-gnu/libc-2.31.so] <- _star
---------------------------------------------------------------------------------------------------------------------------------------------------
  1.6% | 0x18b910 [/usr/lib/x86_64-linux-gnu/libc-2.31.so] <Cnt: 10>
         50.0% |  <- _<std::io::buffered::linewritershim::LineWriterShim<W> as std::io::Write>::wri
                   <- _<std::io::Write::write_fmt::Adapter<T> as core::fmt::Write>::write_str#0x24d4
                   <- core::fmt::num::imp::_<impl core::fmt::Display for i32>::fmt[/home/iipeace/wor
                   <- core::fmt::write[/home/iipeace/work/test/rust]
                   <- _<&std::io::stdio::Stdout as std::io::Write>::write_fmt[/home/iipeace/work/tes
                   <- std::io::stdio::_print[/home/iipeace/work/test/rust] <- rust::main[/home/iipea
                   <- core::ops::function::FnOnce::call_once#0x81d0[/home/iipeace/work/test/rust]
                   <- std::sys_common::backtrace::__rust_begin_short_backtrace[/home/iipeace/work/te
                   <- std::rt::lang_start::_{{closure}}[/home/iipeace/work/test/rust] <- std::rt::la
                   <- std::rt::lang_start[/home/iipeace/work/test/rust] <- main[/home/iipeace/work/t
                   <- __libc_start_main@GLIBC_2.2.5[/usr/lib/x86_64-linux-gnu/libc-2.31.so] <- _star
```



東京 2025
LINUX
PLUMBERS CONFERENCE
TOKYO, JAPAN / DEC. 11-13, 2025

What is Guider?

Function tracing

```
                    runtime.goexit/0x453581 [/home/peacelee/test/go]
                      runtime.main/0x42b14e [/home/peacelee/test/go]
                        main.main/0x48cf39 [/home/peacelee/test/go]
                          fmt.Fprintln/0x48677b [/home/peacelee/test/go]
                            os.(*File).Write/0x482907 [/home/peacelee/test/go]
                              internal/poll.(*FD).Write/0x4822fc [/home/peacelee/test/go]
0.000672 go(919368)             internal/poll.(*FD).writeUnlock/0x4818e0(0x1,0x0,0x0,0x0,0x0,0x0) [/home/peacelee/test/go]
0.010595 go(919367)             runtime.unlock/0x409810(0x0,0x0,0x0,0xc00009bf30,0x26,0xc00009bf38) [/home/peacelee/test/go]
0.010606 go(919367)           runtime.usleep/0x454eb0(0x0,0x11bf48c46887a2,0x0,0xc00009bf30,0x26,0xc00009bf38) [/home/peacelee/test/go]
0.000682 go(919368)               internal/poll.(*fdMutex).rwunlock/0x481740(0x1,0x0,0x0,0x0,0x0,0x0) [/home/peacelee/test/go]
0.000693 go(919368)             runtime.deferreturn/0x428d20(0xc,0x80000000000,0xc0000f006c,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000704 go(919368)           fmt.(*pp).free/0x486400(0xc0000ee008,0x0,0xb,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000715 go(919368)             sync.(*Pool).Put/0x465920(0xc0000ee008,0x0,0xb,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000726 go(919368)               sync.(*Pool).pin/0x465d00(0xc0000ee008,0x0,0xb,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000736 go(919368)                 sync.runtime_procPin/0x436e10(0xc0000ee008,0x0,0xb,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000747 go(919368)               sync.runtime_procUnpin/0x436e40(0xc0000ee008,0x0,0x0,0x0,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000758 go(919368)           runtime.convT64/0x408f30(0xc0000ee008,0x0,0x0,0x0,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000769 go(919368)             runtime.mallocgc/0x40ac10(0xc0000ee008,0x0,0x0,0x0,0x0,0x4,0xfffff80000000001) [/home/peacelee/test/go]
0.000780 go(919368)               runtime.publicationBarrier/0x452da0(0x8,0x0,0x7fe5ba55d008,0xc000366770,0x49f1c0,0xfffff80000000001) [/home/peacele
0.000790 go(919368)           fmt.Fprintln/0x4866f0(0x8,0x4db5e0,0x4d9fe0,0xc000366770,0x49f1c0,0xfffff80000000001) [/home/peacelee/test/go]
                        0x4c40/0x55ff1646ec6e [/usr/bin/yes]
0.024323                    write/0x7f5ea540d1d0(0x1,0x55ff16ddc440,0x1ffe) [/usr/lib/x86_64-linux-gnu/libc-2.31.s
[    rdmem] 0x55ff16ddc440: 'ABCD\nABCD\nABCD\nABCD\nABCD\nABCD\nAB'(32byte)
```

```
__libc_start_main@GLIBC_2.2.5/0x7f0012aca0b3 [/usr/lib/x86_64-linux-gnu/libc-2.31.so]
main/0x40117e [/usr/bin/node]
  node::Start(int, char**)/0x7f00131ef676 [/usr/lib/x86_64-linux-gnu/libnode.so.64]
    node::Start(v8::Isolate*, node::IsolateData*, std::vector<std::__cxx11::basic_string<char, std::char_traits<char>, std::allocato
      node::LoadEnvironment(node::Environment*)/0x7f00131e7f0a [/usr/lib/x86_64-linux-gnu/libnode.so.64]
        0x52cae0/0x7f00131e4b10 [/usr/lib/x86_64-linux-gnu/libnode.so.64]
          v8::Function::Call(v8::Local<v8::Context>, v8::Local<v8::Value>, int, v8::Local<v8::Value>*)/0x7f001344809e [/usr/lib/x86_
            v8::internal::Execution::Call(v8::internal::Isolate*, v8::internal::Handle<v8::internal::Object>, v8::internal::Handle<v
              0xab0720/0x7f0013768762 [/usr/lib/x86_64-linux-gnu/libnode.so.64]
                0xab0110/0x7f001376824e [/usr/lib/x86_64-linux-gnu/libnode.so.64]
                JIT/0x1a739f863ea1 [JIT]
                  Builtin:JSEntryTrampoline/0x1a739f79bd8f [JIT]
                    Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                      Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                        Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                          Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                            Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                              Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                                Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                                  Builtin:InterpreterEntryTrampoline/0x1a739f79ec12 [JIT]
                                    LazyCompile:* /home/peacelee/test/test.js:1/0x1a739f8f7cf1 [JIT]
                                    JIT/0x1a739f8f7a81 [JIT]
                                      0x630d40/0x7f00132e8fb0 [/usr/lib/x86_64-linux-gnu/libnode.so.64]
                                        v8::Function::Call(v8::Local<v8::Context>, v8::Local<v8::Value>, int, v8::Local<v8::Valu
                                          v8::internal::Execution::Call(v8::internal::Isolate*, v8::internal::Handle<v8::interna
                                            0xab0720/0x7f0013768762 [/usr/lib/x86_64-linux-gnu/libnode.so.64]
                                              0xab0110/0x7f001376824e [/usr/lib/x86_64-linux-gnu/libnode.so.64]
                                                JIT/0x1a739f863ea1 [JIT]
                                                  Builtin:JSEntryTrampoline/0x1a739f79bd8f [JIT]
                                                    Builtin:ArgumentsAdaptorTrampoline/0x1a739f7971e3 [JIT]
                                                      LazyCompile:*log console.js:199/0x1a739f8f3411 [JIT]
                                                        LazyCompile:*writeOrBuffer _stream_writable.js:365/0x1a739f8f558e [JIT]
                                                          LazyCompile:*onwriteStateUpdate _stream_writable.js:443/0x1a7
                                                          LazyCompile:*needFinish _stream_writable.js:609/0x1a739f8f7e4
                                                          LazyCompile:*getDefaultTriggerAsyncId internal/async_hooks.js:270
```

```
[Function Leakage Info] [Process: alloc(21499)] [Run: 00:00:03] [Profile: 00:00:02] [Mem: VSS(107.9M)/RSS(105.8M)/PSS(103.3M)/USS(102.4M)] [Leakage: 100.1M]
=============================================================================================================================================
   Size |   Count |     Avg | Function                                                                                                        |
        |    Size |   Backtrace                                                                                                             |
=============================================================================================================================================
 100.0M |       1 |  100.0M | main[/home/iipeace/test/alloc]                                                                                  |
        |  100.0M |  <- 0x29d10[/usr/lib/x86_64-linux-gnu/libc.so.6] <- __libc_start_main@GLIBC_2.2.5[/usr/lib/x86_64-linux-gnu/libc.so.6]
                  <- [.text][/home/iipeace/test/alloc]
---------------------------------------------------------------------------------------------------------------------------------------------
  71.0K |       1 |   71.0K | ??[/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.30]                                                               |
        |   71.0K |  <- 0x63e0[/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2] <- 0x64d0[/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]
                  <- ??[/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2]
---------------------------------------------------------------------------------------------------------------------------------------------
   1.0K |       1 |    1.0K | _IO_file_doallocate@GLIBC_2.3[/usr/lib/x86_64-linux-gnu/libc.so.6]                                              |
        |    1.0K |  <- _IO_doallocbuf@GLIBC_2.3[/usr/lib/x86_64-linux-gnu/libc.so.6]
                  <- _IO_file_overflow@GLIBC_PRIVATE[/usr/lib/x86_64-linux-gnu/libc.so.6]
                  <- _IO_file_xsputn@GLIBC_2.34[/usr/lib/x86_64-linux-gnu/libc.so.6] <- 0x75030[/usr/lib/x86_64-linux-gnu/libc.so.6]
                  <- printf@GLIBC_2.2.5[/usr/lib/x86_64-linux-gnu/libc.so.6] <- main[/home/iipeace/test/alloc]
                  <- 0x29d10[/usr/lib/x86_64-linux-gnu/libc.so.6] <- __libc_start_main@GLIBC_2.2.5[/usr/lib/x86_64-linux-gnu/libc.so.6]
                  <- [.text][/home/iipeace/test/alloc]
---------------------------------------------------------------------------------------------------------------------------------------------

[File Leakage Info] [Process: alloc(21499)] [Run: 00:00:03] [Profile: 00:00:02] [Mem: VSS(107.9M)/RSS(105.8M)/PSS(103.3M)/USS(102.4M)] [Leakage: 100.1M]
=============================================================================================================================================
   Size |   Count |     Avg | Path                                                                                                            |
=============================================================================================================================================
 100.0M |       1 |  100.0M | /home/iipeace/test/alloc                                                                                        |
  71.0K |       1 |   71.0K | /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.30                                                                   |
   1.0K |       1 |    1.0K | /usr/lib/x86_64-linux-gnu/libc.so.6                                                                             |
---------------------------------------------------------------------------------------------------------------------------------------------

[Allocation Histogram] (unit:byte)
=========================================
        Range           |       Count       |
-----------------------------------------
1,024       -     2,047 |          1( 33.3%)|
65,536      -   131,071 |          1( 33.3%)|
67,108,864-134,217,727 |          1( 33.3%)|
-----------------------------------------
```

```
                    <module> [/usr/sbin/iotop:6]
                      main [/usr/lib/python3/dist-packages/iotop/ui.py:680]
                        <lambda> [/usr/lib/python3/dist-packages/iotop/ui.py:727]
                          run_iotop [/usr/lib/python3/dist-packages/iotop/ui.py:615]
                            wrapper [/usr/lib/python3.8/curses/__init__.py:63]
                              run_iotop_window [/usr/lib/python3/dist-packages/iotop/ui.py:601]
                                run [/usr/lib/python3/dist-packages/iotop/ui.py:180]
0.510658 iotop(917437)            resize [/usr/lib/python3/dist-packages/iotop/ui.py:177]
0.510688 iotop(917437)            refresh_processes [/usr/lib/python3/dist-packages/iotop/data.py:459]
0.511301 iotop(917437)              update_process_counts [/usr/lib/python3/dist-packages/iotop/data.py:437]
0.511340 iotop(917437)                list_tgids [/usr/lib/python3/dist-packages/iotop/data.py:404]
0.520806 iotop(917437)                get_process [/usr/lib/python3/dist-packages/iotop/data.py:393]
0.520826 iotop(917437)                  is_monitored [/usr/lib/python3/dist-packages/iotop/data.py:263]
0.520846 iotop(917437)                list_tids [/usr/lib/python3/dist-packages/iotop/data.py:423]
0.520866 iotop(917437)                get_thread [/usr/lib/python3/dist-packages/iotop/data.py:353]
0.520885 iotop(917437)                get_single_task_stats [/usr/lib/python3/dist-packages/iotop/data.py:158]
0.520903 iotop(917437)                  send [/usr/lib/python3/dist-packages/iotop/netlink.py:210]
0.520926 iotop(917437)                    send [/usr/lib/python3/dist-packages/iotop/netlink.py:238]
0.520945 iotop(917437)                      _nl_send [/usr/lib/python3/dist-packages/iotop/netlink.py:25]
```

What is Guider?

File monitoring, Syscall and I/O profiling

[Thread Syscall Info] [Elapsed: 1.137] (Unit: Sec/NR)

Name(TID)	Syscall(SID)	Elapsed	Count	Error	Min	Max	Avg
[TOTAL]							
	pselect6(270)	2.238729	23	0	0.000001	0.882493	0.097336
	pause(34)	1.035837	1	1	1.035837	1.035837	1.035837
	poll(7)	1.030009	7	0	0.000000	0.881513	0.147144
	ppoll(271)	1.029891	11	0	0.000001	0.881969	0.093626
	read(0)	0.001109	26	0	0.000001	0.000280	0.000043
	write(1)	0.000829	27	0	0.000001	0.000107	0.000031
	open(2)	0.000075	2	0	0.000016	0.000059	0.000038
	ioctl(16)	0.000044	5	0	0.000006	0.000012	0.000009
	rt_sigaction(13)	0.000020	30	0	0.000001	0.000004	0.000001
	clock_adjtime(305)	0.000012	4	0	0.000002	0.000005	0.000003
	close(3)	0.000004	2	0	0.000001	0.000003	0.000002
	rt_sigprocmask(14)	0.000004	14	0	0.000000	0.000001	0.000000
	lseek(8)	0.000001	1	0	0.000001	0.000001	0.000001
	rt_sigreturn(15)	0.000000	1	0	0.000000	0.000000	0.000000
	nanosleep(35)	0.000000	1	0	0.000000	0.000000	0.000000
sudo(25532)							
	ppoll(271)	1.029891	11	0	0.000001	0.881969	0.093626
	write(1)	0.000260	7	0	0.000002	0.000095	0.000037
	read(0)	0.000023	7	0	0.000001	0.000008	0.000003
	rt_sigaction(13)	0.000015	28	0	0.000002	0.000002	0.000001
	rt_sigprocmask(14)	0.000004	14	0	0.000000	0.000001	0.000000
screen(909)							
	pselect6(270)	1.135964	17	0	0.000001	0.882493	0.066821
	write(1)	0.000296	11	0	0.000001	0.000101	0.000027
	read(0)	0.000024	6	0	0.000001	0.000007	0.000004

[Thread FS Info] (Unit: NR)

ID	OPT	NrDev	INODE	Size	FS PATH
TOTAL	WRITE			100.0M	
		253:0		100.0M	ext4 / <rw,relatime>
			24520358	100.0M	/home/peacelee/guider/guider/TEST[100.0M]
			0	32.0K	
	READ			147.7M	
		0:3		46.9M	? ?
			0	46.9M	
		253:0		100.8M	ext4 / <rw,relatime>
			24520358	100.0M	/home/peacelee/guider/guider/TEST[100.0M]
guider(932159)	WRITE			100.0M	
		253:0		100.0M	ext4 / <rw,relatime>
			24520358	100.0M	/home/peacelee/guider/guider/TEST[100.0M]
	READ			100.2M	
		0:3		148.0K	? ?
			0	148.0K	
		253:0		100.1M	ext4 / <rw,relatime>
			24520358	100.0M	/home/peacelee/guider/guider/TEST[100.0M]

[Top File Info] [Time: 4793350.080] [Proc: 2] [FD: 0] [File: 15] [CPU: 63%(Usr:54%/Sys:9%)] (Unit: %/MB/NR)

Process (ID / PID / Nr / Pri)	FD	Path
systemd-resolve (597394/ 1/ 1/C 0)	15	SOCKET: 7 EVENT: 6 DEVICE: 1 PROC: 1 PIPE: 0 NORMAL:
	14	anon_inode:[timerfd]
	13	socket:[2622770] (TCP>127.0.0.53:53/LISTEN)
	12	socket:[2622769] (UDP>127.0.0.53:53)
	11	socket:[2622768]
	10	anon_inode:[timerfd]
	9	socket:[2622767]
	8	anon_inode:inotify
	7	/proc/sys/kernel/hostname (0, PATH|NONBLOCK|LARGEFILE|CLOEXEC)
	6	anon_inode:[timerfd]
	5	anon_inode:[signalfd]
	4	anon_inode:[eventpoll]
	3	socket:[2622765]
	2	socket:[2620882]
	1	socket:[2620882]
	0	/dev/null (0, DIRECTORY|LARGEFILE)
tmux: server (505691/ 1/ 1/C 0)	9	DEVICE: 6 PIPE: 2 SOCKET: 1 EVENT: 0 NORMAL: 0 PROC:
	10	/dev/ptmx (0, RDWR|NONBLOCK|LARGEFILE)
	9	/dev/ptmx (0, RDWR|NONBLOCK|LARGEFILE)
	8	/dev/ptmx (0, RDWR|NONBLOCK|LARGEFILE)
	6	socket:[1585533] (UDS>/tmp/tmux-1001/default)
	4	pipe:[1585532]

東京 2025
LINUX
PLUMBERS CONFERENCE
TOKYO, JAPAN / DEC. 11-13, 2025

Real-Time Monitoring & Automation

Continuous system behavior monitoring

- Guider monitors many things from boot

Thresholds defined via JSON — flexible configuration

- Target item, value (threshold), handling commands, ...

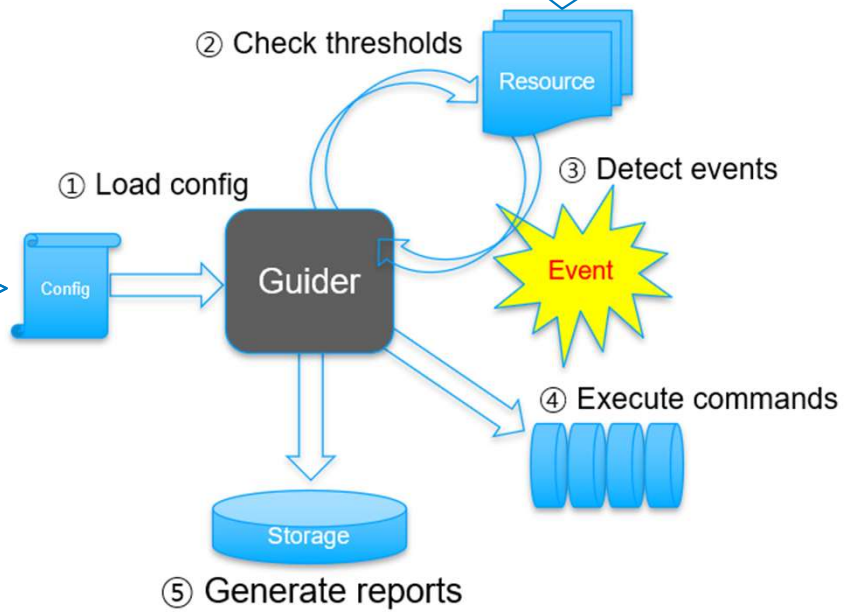
Type	Items
Device	CPU, GPU, RAM, Block, Storage, Network, Battery, Time
Logical	Swap, Load, Fd, Socket, File, Task, PSI, Cgroup
Log	DLT, Kernel, Journal, Ftrace, Android
Function	C, C++, Rust, Go, Java, Javascript, Python, Syscall
IPC	D-Bus, Signal, Binder

Format

```
"Target": {
    "Scope": [
      {
        "Apply Flag",
        "Threshold",
        "Duration Condition",
        "Time Condition",
        "Task Monitoring",
        "Detection Count",
        "Reactivation Time",
        "Command List",
      },
    ],
...
```

Example

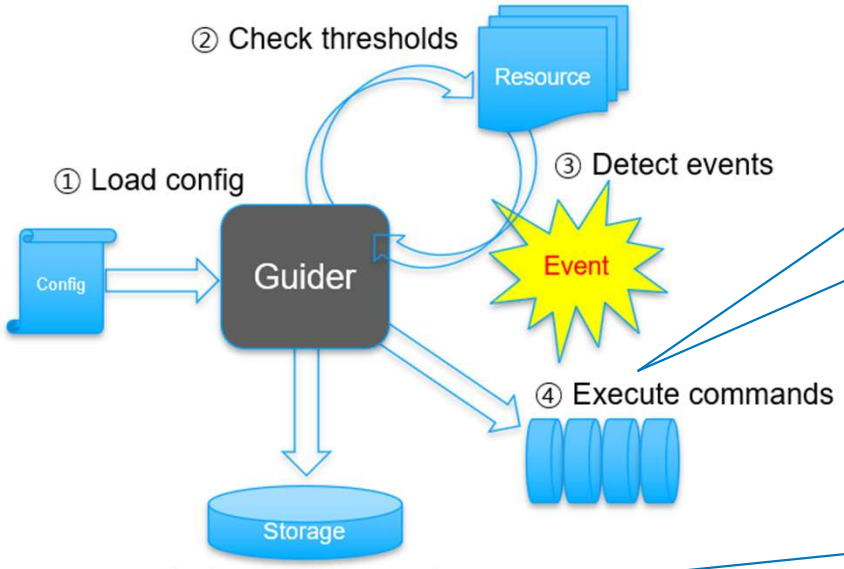
```
"cpu":{
    "SYSTEM": [
      {
        "apply": "true",
        "total": 95,
        "interval": 5,
        "after": "60s",
        "taskmon": "true",
        "oneshot": "true",
        "refresh": 10,
        "command": ["SAVE:5s", "CMD_FUNCREC"],
      },
    ],
    "yes*": [
      {
      ...
```



Real-Time Monitoring & Automation

Autonomous generation of diagnostic reports when issues occur

- Includes past context (pre-trigger trace buffers)




```
"COMMAND": {

    "CMD_NOTIFY_WARN": "GUIDER cli upstream:notify:WARN",
    "CMD_TOP": "GUIDER top -o OUTDIR/ -R 10 -Y c:15 -q REPFILEFORM:EVTNAME",
    "CMD_FTOP": "GUIDER ftop -o OUTDIR/guider_top_EVTNAME_EVTTIME.out -a -R 1 -Y c:15",
    "CMD_FTOP_PROC": "GUIDER ftop -o OUTDIR/guider_top_EVTNAME_EVTTIME.out -a -g EVTPID -R 1 -Y c:15",
    "CMD_MTOP": "GUIDER mtop -o OUTDIR/ -R 3 -Y c:15 -q REPFILEFORM:EVTNAME",
    "CMD_FUNCREC": "GUIDER funcrec -s OUTDIR/ -R 3 -q REPFILEFORM:EVTNAME",
    "CMD_DISKTOP": "GUIDER disktop -R 3 -a -o OUTDIR/ -Y c:15 -q REPFILEFORM:EVTNAME",
    "CMD_DISKTOP_TOTAL": "GUIDER disktop -e T -R 3 -a -o OUTDIR/ -Y c:15 -q REPFILEFORM:EVTNAME",
    "CMD_NETTOP": "GUIDER ntop -R 3 -o OUTDIR/ -Y c:15 -q REPFILEFORM:EVTNAME",
    "CMD_TTOP_PROC": "GUIDER ttop -g EVTPID -P -R 3 -o OUTDIR/ -Y c:15 -q REPFILEFORM:EVTNAME",
    "CMD_TTOP_CTX": "GUIDER ttop -S C:1 -R 2 -o OUTDIR/ -Y c:15 -q REPFILEFORM:EVTNAME",
    "CMD_TTOP_UTOP_ALL": "GUIDER ttop -g EVTPID -P -S c:1 -e E -R 1 -Q -c \"GUIDER utop -g PID -H -o OUTDIR/ -R 5 -q LAZYCACHE,
    "CMD_TTOP_UTOP_2": "GUIDER ttop -g EVTPID -P -S c:1 -e E -R 1 -Q -q NRTOPRANK:2 -c \"GUIDER utop -g PID -H -o OUTDIR/ -R 5 -
    "CMD_UTOP": "GUIDER utop -g EVTPID -H -o OUTDIR/ -R 3 -q REPFILEFORM:EVTNAME",
    "CMD_LEAK_1G": "GUIDER leaktrace -g EVTPID -T OUTDIR/libleaktracer.so -o OUTDIR/ -c 1g -q REPFILEFORM:EVTNAME",
    "CMD_LEAK_+200M": "GUIDER leaktrace -g EVTPID -T OUTDIR/libleaktracer.so -o OUTDIR/ -c +200m -q REPFILEFORM:EVTNAME",
    "CMD_SPERF": "GUIDER sperf -o OUTDIR -R 2 -Y i:0 -q REPFILEFORM:EVTNAME",
    "CMD_SPERF_PROC": "GUIDER sperf -g EVTPID -o OUTDIR -R 2 -Y i:0 -q REPFILEFORM:EVTNAME",
    "CMD_HPROF_PROC": "GUIDER hprof -g EVTPID -o OUTDIR -R 10 -Y i:0 -q REPFILEFORM:EVTNAME",
    "CMD_HPROF_PROC_+50M": "GUIDER hprof -g EVTPID -o OUTDIR -Y i:0 -q STOPRSS:+50M, REPFILEFORM:EVTNAME"

},
```

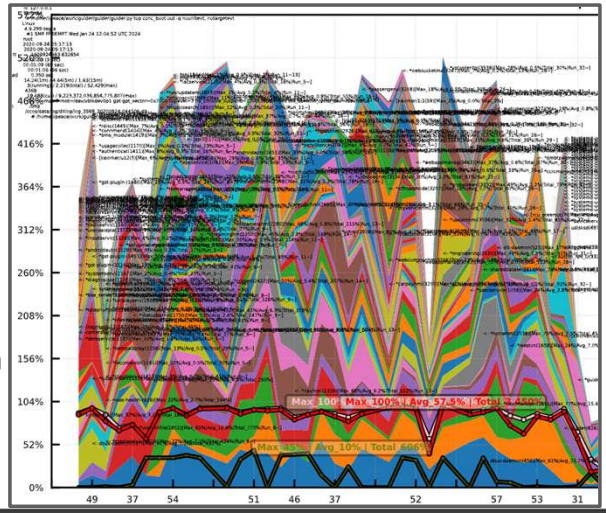
```
root@mst6g:~# ls /ccos/data/log/guider/
guider_00000000_00000001_CPU_SYSTEM_total_90_SAVERAW_2020-09-22T12:59:04Z_47469.out.gz
guider_00000000_00000002_CPU_SYSTEM_total_90_SAVERAW_2020-09-22T13:14:02Z_48128.out.gz
guider_00000000_00000002_CPU_SYSTEM_total_90_SAVERAW_2020-09-22T13:14:02Z_48128.out.gz.gz
guider_00000000_00000003_CPU_SYSTEM_total_90_SAVERAW_2020-09-22T13:14:32Z_48150.out.gz
guider_00000001_00000001_CPU_SYSTEM_total_80_SAVERAW_2020-09-24T09:54:19Z_5152.out.gz
guider_00000002_00000001_CPU_SYSTEM_total_95_SAVEMIN:-2_2020-10-26T13:25:52Z_2500403.out.gz
guider_00000002_00000002_CPU_SYSTEM_total_95_SAVEMIN:-2_2020-10-26T13:26:03Z_2500460.out.gz
guider_00000003_00000001_CPU_SYSTEM_total_95_SAVEMIN:-2s_2020-10-26T13:28:33Z_2501192.out.gz
guider_00000004_00000001_CPU_SYSTEM_total_95_SAVEMIN:-2s_2020-10-26T13:28:56Z_2501285.out
guider_00000005_00000001_USER_SAVE_2020-10-26T13:35:26Z_2503149.out.gz
guider_00000005_00000002_USER_SAVEMIN_2020-10-26T13:35:39Z_2503209.out.gz
guider_00000005_00000003_USER_SAVERAW_2020-10-26T13:35:49Z_2503268.out.gz
```

Real-Time Monitoring & Automation

Autonomous generation of diagnostic reports when issues occur

- Includes past context (pre-trigger trace buffers)

Visualization



```
[Top Info] [Time:  53.290] [Inter: 1.3] [Ctxt: 30021] [Life: +112/-46] [IRQ: 21379] [Core: 6] [Task: 349/1952] [Load: 6/1/0] [RAM: 6.5G] [Swap: 2.3G]
           [PSI > [cpu] some(56.4%) [io] some(3.0%) / full(0.7%) [memory] some(0%) / full(0%)]

 ID   | CPU(Usr/Ker/Blk/IRQ)|MemAvl(Per/  User/ Cache/ Kern)| Swap( Per/ In/Out)|  PgRclm   | BlkRW | NrFlt |Blk| NrSIRQ | PgMlk | PgDirt |   NetIO    |
Total  | 92 %( 67/ 22/  0/  0)|  4376( 66/  1200/  1174/  546)|    0(   0/  0/  0)|     0/0   | 3/10  |    4  | 0 |     0  |    0  |     5  |  1K/3K     |
NVIDIA/18000000.vgpu    (  0 %)|

           Process ( PID/PPID/  Nr/ Pri)| CPU(Usr/Ker/Dly)| VSS( RSS/Txt/Shr/Swp)| Blk(   RD/   WR/NrFlt
      dbus-daemon ( 444/    1/   1/C-19)|  50( 30/ 20/  -)|   7(   5/  0/  2/  0)|   0(   -/   -/   0
          homehmi ( 2354/   1/  23/R 80)|  50( 39/ 10/  -)|1749( 229/  1/120/  0)|   0(   -/   -/   0
  globalsearchhmi ( 3198/   1/  12/C  0)|  34( 29/  5/  -)| 726(  82/  2/ 60/  0)|   0(   -/   -/   1
  engineeringmode ( 3197/   1/  15/C  0)|  33( 26/  6/  -)|1167(  80/  8/ 68/  0)|   0(   -/   -/   0
  vehicleupdatehm ( 3202/   1/  22/C  0)|  26( 21/  5/  -)|1222(  63/  2/ 52/  0)|   0(   -/   -/   0
  shareddataservi ( 852/    1/   5/C  0)|  16( 12/  3/  -)| 319(  10/  0/  8/  0)|   0(   -/   -/   0
   vehicleservice ( 415/    1/  13/C  0)|  14( 12/  2/  -)|1112(  24/  9/ 16/  0)|   0(   -/   -/   0
       dlt-daemon ( 517/    1/   1/C  0)|  13(  4/  9/  -)|  12(  11/  0/  1/  0)|   0(   -/   -/   0
  retrievermanage ( 1789/   1/  42/C  0)|  13(  9/  3/  -)| 690(  54/  1/ 29/  0)|   0(   -/   -/   0
          updatehmi ( 3201/ 1/  17/C  0)|  13( 10/  3/  -)| 901(  46/  2/ 41/  0)|   2(   -/   -/   0
  quickcontrolhmi ( 3199/   1/  12/C  0)|  12( 10/  2/  -)| 692(  48/  1/ 43/  0)|   0(   -/   -/   0
      userservice ( 1032/   1/   5/C  0)|   9(  7/  1/  -)| 294(  15/  0/ 12/  0)|   1(   -/   -/   0
      dbus-daemon ( 551/    1/   1/C  0)|   8(  6/  2/  -)|   6(   4/  0/  2/  0)|   0(   -/   -/   0
       climatehmi ( 3196/   1/  13/C  0)|   8(  6/  2/  -)| 483(  44/  2/ 39/  0)|   0(   -/   -/   0
  locationsharing ( 2957/   1/  19/C  0)|   6(  3/  2/  -)| 783(  83/ 15/ 62/  0)|   0(   -/   -/   0
          navihmi ( 3127/   1/  20/R 80)|   6(  5/  0/  -)| 632(  76/  0/ 63/  0)|   0(   -/   -/   0
  vrcontentplayer ( 3203/   1/  12/C  0)|   5(  3/  1/  -)| 665(  42/  1/ 37/  0)|   0(   -/   -/   0
      powerservice ( 247/   1/  16/C  0)|   3(  2/  0/  -)|1053(  13/  1/ 11/  0)|   -/   -/   0
  systemd-journal ( 474/    1/   1/C  0)|   3(  2/  1/  -)|  31(  15/  0/ 14/  0)|   -/   -/   0
  node-health-mon ( 897/    1/   6/C  0)|   3(  2/  0/  -)| 314(  12/  0/  5/  0)|   -/   -/   0
      appservice (1063/     1/   6/C  0)|   3(  2/  0/  -)| 344(  14/  0/ 11/  0)|   -/   -/   0
  locationservice (1121/    1/  20/C  0)|   3(  1/  1/  -)|1542(  22/  2/ 18/  0)|   -/   -/   0
```

Details

```
[Top Summary Info]

 IDX |        Interval        | CPU |       Avl        | BlkRW | Blk | SWAP | NrPgRclm | NrFlt | NrCtx | NrIRQ |  NrTask  | Core |  NetIO
   1 |  51.990 -  53.290 |  92 | 4376/1200/1174 |  3/10 |   0 |    0 |      0/0 |     4 | 30021 | 21379 | 349/1952 |    6 | 1K/3K
   2 |  53.290 -  54.350 |  60 | 4290/1237/1188 |  10/1 |   0 |    0 |      0/0 |     8 | 20295 | 15582 | 349/1969 |    6 | 1K/3K
   3 |  54.350 -  55.400 |  66 | 4246/1270/1295 | 102/0 |   0 |    0 |      0/0 |    74 | 20358 | 16316 | 384/2025 |    6 | 1K/1K
   4 |  55.400 -  56.440 |  83 | 4233/1270/1314 |  25/0 |   0 |    0 |      0/0 |     6 | 20845 | 16073 | 383/2025 |    6 | 1K/1K
   5 |  56.440 -  57.490 |  17 | 4228/1282/1318 |   0/0 |   0 |    0 |      0/0 |     2 | 10130 |  7753 | 383/2024 |    6 | 1K/1K

[Top CPU Info] (Unit: %) (New: +) (Die: -)

      COMM (   PID/  PPID/  Nr/ Pri)| Min/Avg/Max/Tot |   1   2   3   4   5
   [CPU/AVG] (   -/    -/   -/  -)| 17/63.6/92/318   |  92  60  66  83  17
  globalsearch (  3198/    1/  12/C  0)| 13/26.0/83/130   |  34  83  13   0   0
  dbus-daemon (   444/    1/   1/C-19)|  5/21.8/50/109   |  50  30  19   5   5
  systemd-udev (  594/    1/   1/C  0)|  0/17.0/65/85    |   0   0  20  65   0
  dbus-daemon (   551/    1/   1/C  0)|  3/14.2/40/71    |   8   3  20  40   0
  quickcontrol (  3199/   1/  17/C  0)| 12/11.8/47/59    |  12  47   0   0   0
      homehmi (   2354/   1/  23/C  0)|  1/10.6/50/53    |  50   2   0   1   0
```

Other Info

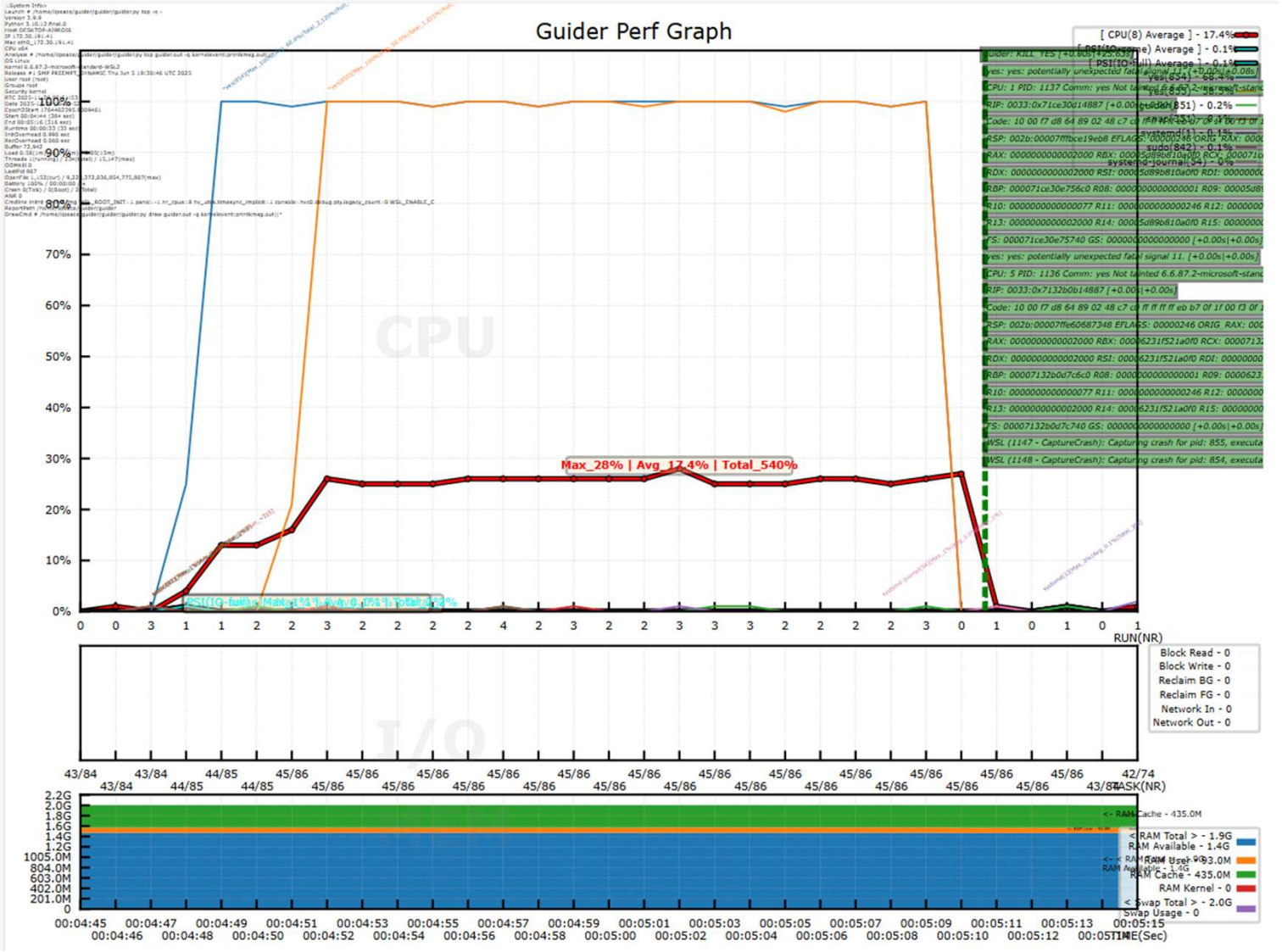
```
[Top Delay Info] [Top Prio Info]
[Top GPU Info] [Top VSS Info]
[Top RSS Info] [Top Block Info]
[Top Storage Info] [Top Network Info]
[Top Cgroup.CPU Info] [Top Cgroup.Throttle Info]
[Top Cgroup.Mem Info] [Top Cgroup.Blk Info]
```

東京 2025
Linux Plumbers Conference
TOKYO, JAPAN / DEC. 11-13, 2025

Real-Time Monitoring & Automation

Supporting various logs and visualization

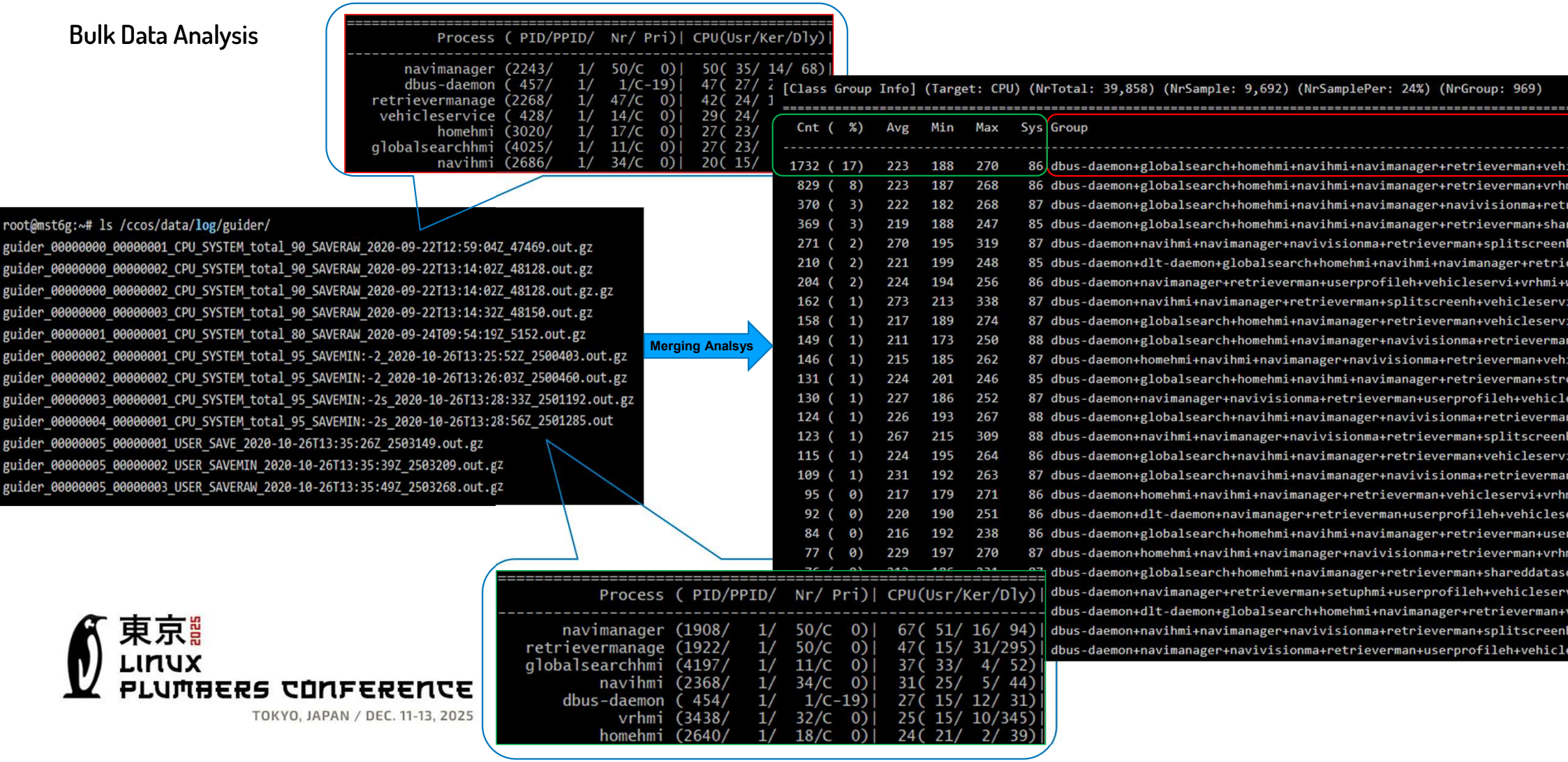
- Kernel
- Journal
- Ftrace
- DLT
- Android (logcat)
- ATRACE



東京 2025
LINUX
PLUMBERS CONFERENCE
TOKYO, JAPAN / DEC. 11-13, 2025

Real-Time Monitoring & Automation

Bulk Data Analysis



東京 2025
LINUX PLUMBERS CONFERENCE
TOKYO, JAPAN / DEC. 11-13, 2025

Remote APIs

Oneshot

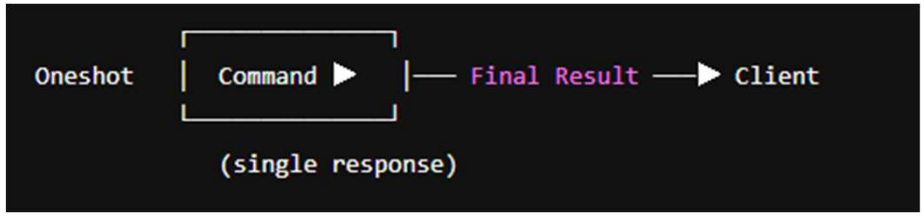
- Receives only the final result of a command execution
- Exam) show all system info

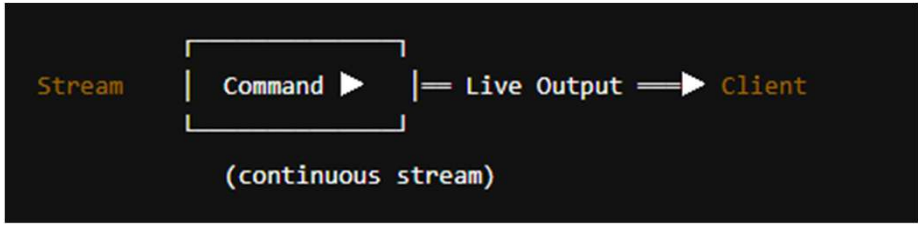
Stream

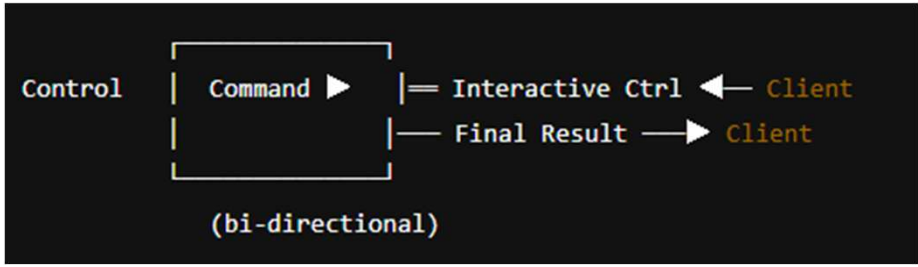
- Continuously receives real-time output produced during command execution
- Exam) show real-time system performance data

Control

- Allows the client to send additional control commands while a task is running, and then receive the final result when the operation completes.
- Exam) run perf or perfetto tools



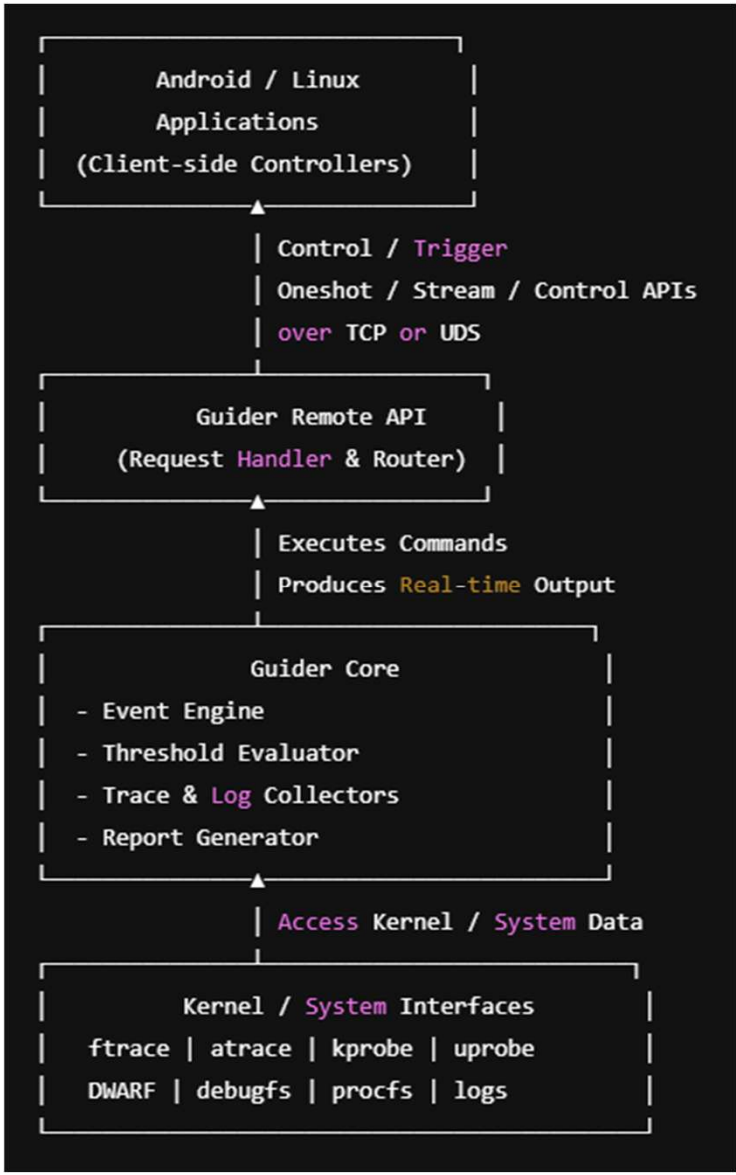




Remote APIs

Making Android/Linux Applications using Guider's remote APIs based on TCP/UDS

1. Application sends a command or subscribes to events
2. Remote API Layer receives request (Oneshot/Stream/Control)
3. Guider Core executes monitoring/profiling logic
4. Kernel interfaces provide actual tracing & perf data
5. Guider Core returns results
6. Application consumes data and visualizes or triggers next action

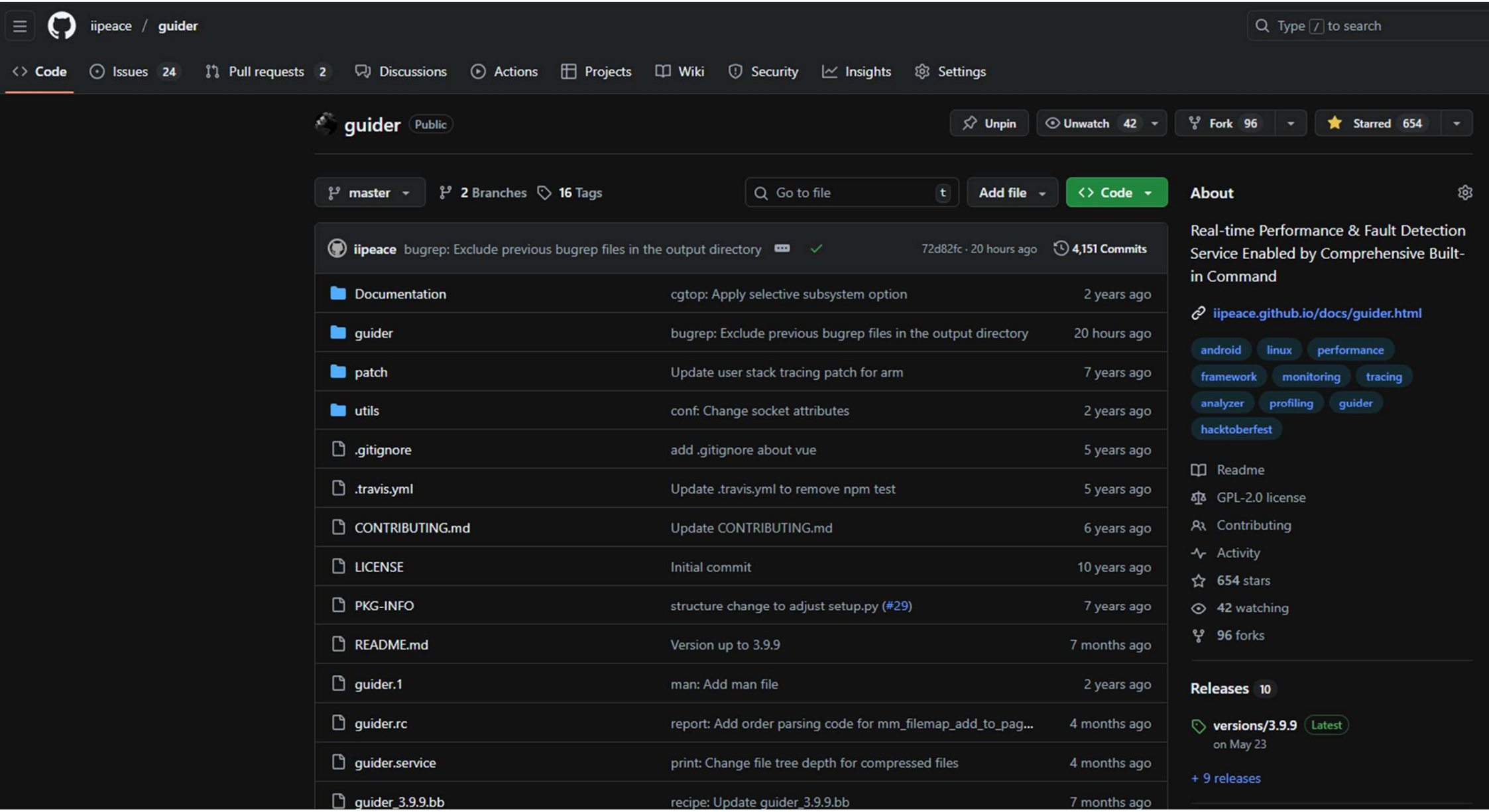


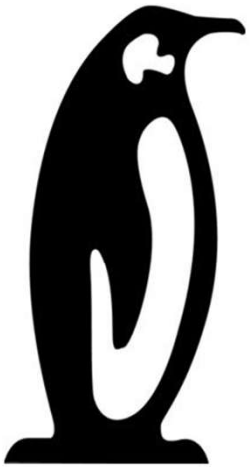
Remote APIs

Building higher-level applications using Guider's APIs

1. Real-time resource monitoring
2. Resource history analysis
3. Function profiling capabilities
4. Bug-recording feature
5. Running system features




iipeace / guider
Type / to search
Code
Issues 24
Pull requests 2
Discussions
Actions
Projects
Wiki
Security
Insights
Settings
guider Public
Unpin
Unwatch 42
Fork 96
Starred 654
master
2 Branches
16 Tags
Go to file t
Add file
Code
iipeace bugrep: Exclude previous bugrep files in the output directory
72d82fc · 20 hours ago
4,151 Commits
Documentation cgtop: Apply selective subsystem option 2 years ago
guider bugrep: Exclude previous bugrep files in the output directory 20 hours ago
patch Update user stack tracing patch for arm 7 years ago
utils conf: Change socket attributes 2 years ago
.gitignore add .gitignore about vue 5 years ago
.travis.yml Update .travis.yml to remove npm test 5 years ago
CONTRIBUTING.md Update CONTRIBUTING.md 6 years ago
LICENSE Initial commit 10 years ago
PKG-INFO structure change to adjust setup.py (#29) 7 years ago
README.md Version up to 3.9.9 7 months ago
guider.1 man: Add man file 2 years ago
guider.rc report: Add order parsing code for mm_filemap_add_to_pag... 4 months ago
guider.service print: Change file tree depth for compressed files 4 months ago
guider_3.9.9.bb recipe: Update guider_3.9.9.bb 7 months ago
About
Real-time Performance & Fault Detection Service Enabled by Comprehensive Built-in Command
iipeace.github.io/docs/guider.html
android
linux
performance
framework
monitoring
tracing
analyzer
profiling
guider
hacktoberfest
Readme
GPL-2.0 license
Contributing
Activity
654 stars
42 watching
96 forks
Releases 10
versions/3.9.9 Latest
on May 23
+ 9 releases



東京 2025
LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DECEMBER 11-13, 2025

